Detailed Solutions for Case Studies 1–5

Case Study 1 – Employee Salaries

Data: 4, 5, 5, 6, 7, 8, 8, 9, 30

1. Mean, Median and Mode

Formula:
Mean = $\Sigma x / n$
Median = Middle value
Mode = Most frequent value

Calculation:
$\Sigma x = 4+5+5+6+7+8+8+9+30 = 82$
$n = 9$
Mean = 82/9 = 9.11 lakhs
Median = 7
Mode = 5 and 8

2. Range

Formula: Range = Maximum − Minimum
Range = 30 − 4 = 26

3. Variance & Standard Deviation

Formula:
Variance (Population) = $\Sigma(x-\mu)^2 / n$
Standard Deviation = $\sqrt{\text{Variance}}$

Result:
Variance = 56.99
Standard Deviation = 7.55

4. Outlier

The value 30 is much larger than the remaining salaries and is an outlier.

5. Conclusion

Median (7 lakhs) best represents the typical salary because the outlier increases the mean.

Case Study 2 – Website Response Time

Data: 120,125,128,130,132,135,140,145,150,300

1. Mean & Median

Formula:
Mean = $\Sigma x/n$
Median = Average of middle two values

$\Sigma x = 1505$
Mean = 1505/10 = 150.5 ms
Median = (132+135)/2 = 133.5 ms

2. Range

Range = 300−120 = 180 ms

3. Variance & Standard Deviation

Variance = $\Sigma(x-\mu)^2/n$
Standard Deviation = $\sqrt{\text{Variance}}$
Variance = 2558.05
Standard Deviation = 50.58 ms

4. Skewness

Since one value (300) is very high, the data is positively (right) skewed.

5. Conclusion

Median is more reliable because it is not affected by the extreme response time.

Case Study 3 – Student Performance

Data: 42,45,48,50,52,52,55,58,60,62,65,95

1. Quartiles

$Q1 = 49$
$Q2$ (Median)$=53.5$
$Q3=61$

2. IQR

Formula: $IQR = Q3 - Q1$
$IQR = 61-49 = 12$

3. Outlier

Lower Limit $= Q1-1.5\times IQR = 49-18 = 31$
Upper Limit $= Q3+1.5\times IQR = 61+18=79$
$95>79$, therefore 95 is an outlier.

4. Mean & Median

Mean = 57
Median = 53.5

5. Conclusion

The outlier (95) increases the mean but has very little effect on the median.

Case Study 4 – E-Commerce Orders

Data: 18,20,22,22,24,25,26,28,30,35

1. Mean, Median & Mode

$\Sigma x=250$
Mean=250/10=25
Median=(24+25)/2=24.5
Mode=22

2. Variance & Standard Deviation

Variance = 22.8
Standard Deviation = 4.77

3. Coefficient of Variation

Formula: $CV=(SD/Mean)\times 100$
$=(4.77/25)\times 100=19.1\%$

4. Interpretation

Orders vary by about 4.77 orders from the average.

5. Conclusion

CV of 19.1% indicates moderate variability.

Case Study 5 – Comparing Two Data Science Models

Model A: 2,3,3,4,4,5,5,6
Model B: 1,1,2,4,5,7,8,9

1. Mean Error

Formula: $Mean = \Sigma x/n$
Model A: $32/8 = 4$
Model B: $37/8 = 4.625$

2. Variance

Formula: $Variance = \Sigma(x - \mu)^2/n$
Model A=1.50
Model B=9.23

3. Standard Deviation

Formula: $SD = \sqrt{Variance}$
Model A=1.22
Model B=3.04

4. Consistency

Model A is more consistent because it has the smaller standard deviation.

5. Conclusion

A lower mean error alone does not guarantee a better model. Lower variability is also important.